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Engineering of Innovative Technologies: User research

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1 Project description

2 Persona

Profile

- Name: Søren
- Age: 46
- Occupation: Surgical doctor
- Experience: 18 years in surgery

Goal & Motivation

- Ensure optimal patient care during critical situations
- Develop your team's coordination and communication
- Keep up to date with modern training methods
- Reduce stress and errors in acute situations
- Build trust and security in the team

2.1 Story

Søren has worked in the surgical department for 18 years and has seen the development from paper-based systems to digital solutions. He is passionate about patient care, but frustrated by communication errors in pressured situations. He has experienced how lack of team coordination can negatively affect patient outcomes. Søren is open to technology, but requires that it add real value to the treatment.

2.2 Frustrations

- **Communication errors:** Misunderstandings between team members during critical procedures
- **Training restrictions:** Limited opportunities to practice realistic emergency scenarios
- **Standardization:** Varying competency levels among new employees
- **Resource constraints:** Lack of dedicated training facilities

3 The Customer Profile

Gains

- **Improved patient care:** Faster and more accurate decisions
- **Teamwork:** Better communication and coordination
- **Competence development:** Standardized and measurable training
- **Safety:** Increased confidence in acute situations
- **Efficiency:** Reduced training time with higher quality
- **Innovation:** Recognized as a leader in modern education

Pains

- High implementation costs and budget constraints
- Technical complexity and integration challenges
- Employee resistance to new technologies
- Time-consuming training in already busy workdays
- Measuring impact - how to prove ROI and clinical impact
- Maintenance and ongoing technical support needs

3.1 Customer jobs

- **Functional:** Train teams in emergency procedures and communication
- **Emotional:** Create safety and confidence in employees
- **Social:** Be recognized for innovative training methods

4 User Research Process

4.1 Problem Hypothesis

We hypothesize that novice clinical staff and trainees experience coordination failures in acute, multi-person events primarily because of:

1. Ambiguous role and ownership during overlapping tasks; and
2. Lack of short, pragmatic decision aids for quickly evolving complications; and
3. Low perceived fidelity of current simulation training which limits stress-exposure and adaptive practice.

These factors reduce decision speed and confidence in time-critical moments (e.g., trauma calls, unexpected intra-operative bleeding) and increase dependency on senior staff rather than reliable team processes.

Primary success criteria (how we will know the hypothesis is true):

- Faster and more consistent time-to-priority decisions in pilot simulations (median reduction expected).
- Improved trainee self-rated confidence (pre/post Likert increase) when left to perform initial actions.
- Fewer coordination conflicts (measured as count of conflicting orders or role-claims) during micro-simulations.

4.2 Interviews

We began with three in-depth interviews from our early-stage user research. The interviewees were:

- Jakob — medical student, 25
- Sebastian — medical student, 26
- Simone — nurse, 28

4.3 Where & When

Locations:

- Emergency departments at regional hospitals
- Online interviews (Teams/Zoom)

Target identification:

- Employees with name tags from acute/intensive care units
- Participants in emergency medicine courses
- People in doctor/nurse uniforms on departments

Timing:

- Break periods (lunch 12-13, coffee 10-10:30, 15-15:30)
- Shift periods (7-8, 15-16, 23-24)
- After professional meetings and conferences
- Scheduled interview appointments

4.4 How do you approach your target customer

Medium of contact:

- Direct contact at departments (with management permission)
- University with medicine students

Introduction: "Hi [Name], my name is [name] and I come from SDU. We develop innovative training solutions for the healthcare sector and would like to learn from your experience with team training in acute situations. Do you have 10-15 minutes to share your experiences? We are not selling anything — we are only here to learn."

Attire: Professional business casual Visible ID card or identification Respectful of hospital environment and hygiene guidelines

4.5 Interview

Intro questions:

1. "What is your age?"
2. "What is your profession?"
3. "What is your experience?"

Key questions:

1. "Do you have problems with team coordination in emergencies?"
2. "Tell me about the last time you had a critical situation where communication was challenging."
"Why was it challenging?"
3. "In which situations do you experience these problems the most?"

In-depth questions:

- "Tell me about the last time you had to train in pressure situations that required collaboration"
- "Did you experience any difficulties as part of that training?"
"What was the most difficult part, and why?"

- "How do you train team coordination training now?"
- "What are the positives or negatives of the current solution?"
- "Do you experience stressful situations in your field when working collaboratively with others?"
- "What was your biggest concern the last time you were in a stressful situation?"

Interview principles:

- 80% listening, 20% asking
- Focus only on the past: "Have you ever..." or "Tell me about the last time..."
- Use "why" as the ultimate weapon
- Don't talk about solutions - only about problems

User Journey Mapping:

- Map a typical emergency situation from alarm to closure
- Identify touchpoints where coordination is critical
- Map emotions and frustrations throughout the process

Observational studies:

- Observe training scenarios at simulation centers
- Follow teams during emergency situations (with permission)
- Observe communication patterns and coordination

Expert interviews:

- Education managers at hospitals
- Simulation instructors
- Quality managers and patient safety specialists
- IT managers with experience in health technology

Desk Research:

- Danish patient safety reports
- International studies on team training in healthcare
- Competitor analysis of existing training solutions
- Industry reports and market analyses

4.6 Synthesis of current interviews

From the three interviews we distilled the following cross-cutting insights (representative interview references in parentheses):

1. Ambiguity of role/ownership in multi-team events is a common failure mode — trainees report difficulty when multiple teams interact (Jacob).
2. Surprise events (unexpected bleeding, sudden deterioration) rapidly change task priorities and trainees want short, actionable checklists to guide the first actions (Sebastian).
3. Novice staff are uncertain about escalation thresholds and when to act independently vs call for mentor support; morning huddles help but don't remove the ambiguity (Simone).
4. Existing simulations and workshops are accessible and useful for procedural practice but often lack the realism and multi-person stress exposure needed to practice prioritization and coordination (Jacob, Sebastian).
5. Psychological stress is focused on personal performance fear ("not to f*** up") which reduces willingness to take early decisive actions without senior support (Jacob, Simone).

These patterns validate parts of the Problem Hypothesis and identify candidate interventions that are low-cost to pilot.

4.7 Prioritized problems

Ranked by cross-interview frequency and safety impact:

1. Role and ownership ambiguity during multi-team events (high frequency for trainees in trauma/operating settings) — Jacob explicitly called this out.
2. Lack of short practical decision aids (one-page checklists or prioritization trees) for surprise events such as unexpected bleeding — Sebastian's primary ask.
3. Simulation fidelity and stress-exposure gap: simulations often do not recreate overlapping team dynamics and time pressure (Jacob + Sebastian).
4. Escalation threshold uncertainty for novice staff — Simone highlighted when to call for help vs act.